Information theory: a more general perspective on the brain

Computational Neuroscience 2026/27 · Lesson 0

Davide Orsenigo

Wednesday 4 November 2026, room G

Following Cover & Thomas, *Elements of Information Theory* (Wiley 2006), ch. 2

Outline

Today: three questions and three labs

Opening	Does a correlation see every dependence?	6 min
Part I	How uncertain is one signal? Entropy	16 min
Lab A	Notebook, Parts 1–2: the entropy of one region and of a pair	15 min
Part II	How much does one signal tell about another? Mutual information	18 min
Lab B	Notebook, Parts 3–4: mutual information, its bias, comparison with the FC	15 min
Part III	Can three signals share what no pair shows?	14 min
Lab C	Notebook, Part 5: copies, XOR and triplets of brain regions	13 min
Discussion	What this means for whole-brain models, the topic of the next lesson	8 min

The data, and what we usually compute from them

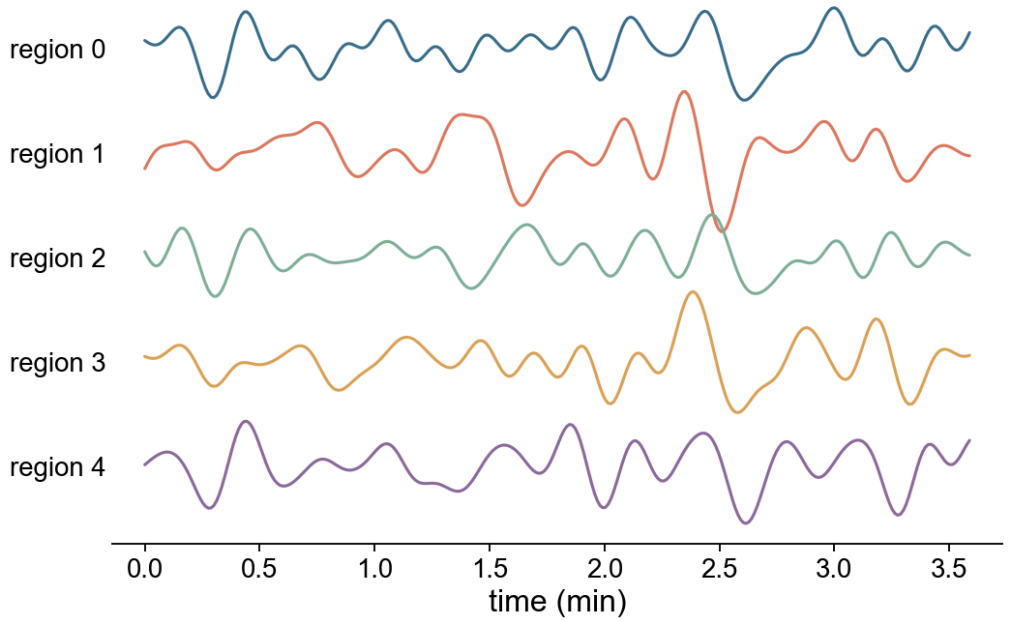
One resting-state fMRI recording of the Human Connectome Project: 1200 **volumes** (brain images), one every 0.72 s, 14.4 minutes at rest.

In each volume the BOLD signal is averaged over each of 200 cortical **regions** (the parcels of the Schaefer atlas): 200 signals of 1200 values.

The usual summary is the **functional connectivity** (FC): for each pair of regions, the Pearson correlation ρ of their two signals. Whole-brain models are fitted to it (next lesson).

Question

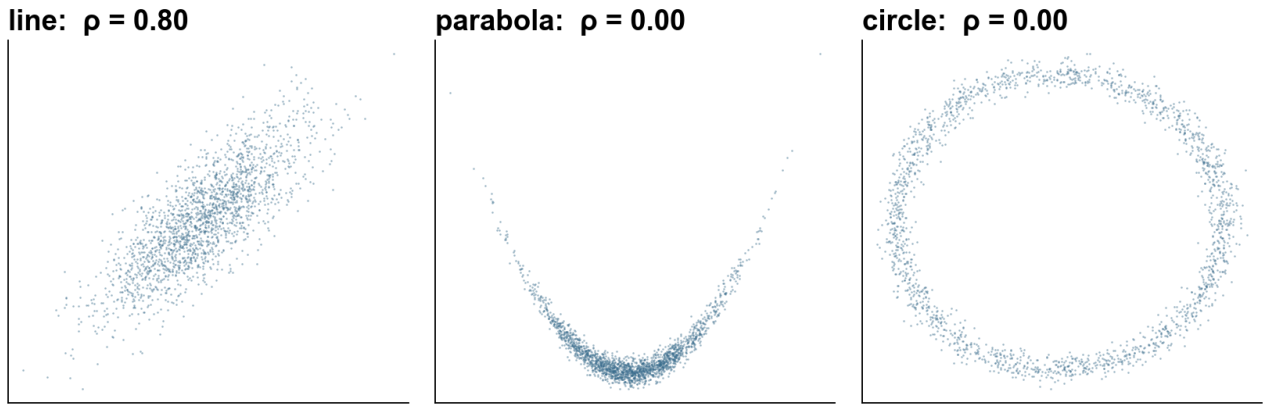
Does a correlation see every way in which two regions can depend on each other?



Five of the 200 regions, z-scored: each signal shifted to mean 0 and scaled to standard deviation 1. The first 3.6 minutes.

Data: HCP (Van Essen et al. 2013); Schaefer atlas (Schaefer et al. 2018)

Correlation: how well a straight line fits



$$\rho = \langle \tilde{x}\,\tilde{y} \rangle, \qquad \tilde{x} = \frac{x - \bar{x}}{\sigma_x}, \;\; \tilde{y} = \frac{y - \bar{y}}{\sigma_y}$$

The average product of the z-scored signals: ± 1 when the points lie on a straight line, 0 when positive and negative products cancel.

Parabola: x is symmetric around 0 and y is fixed by x up to the noise. A point at x and one at $-x$ have the same y: their products cancel, $\rho = 0$.

So

$\rho = 0$ does not mean that the signals are **independent**, that is, that knowing one tells nothing about the other.

Knowing one tells something about the other when it reduces our uncertainty about it. So we first need a measure of uncertainty.

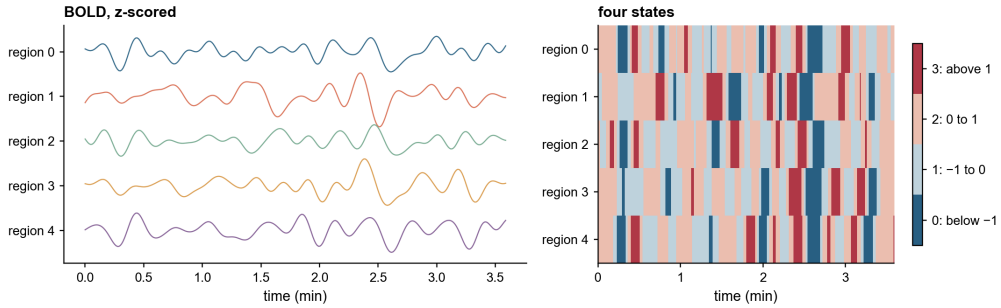
20,000 samples per cloud (notebook, Part 3). The circle: $\rho = 0$ by the same symmetry

How uncertain is one signal? Entropy

Shannon (1948) · Cover & Thomas §2.1–2.2

From a signal to symbols

1. We estimate the **probabilities of states** by counting, without assuming their shape. A continuous signal never takes the same value twice, so we cut its range into bins and count how often it falls in each.
2. Four bins, with edges at −1, 0 and +1 standard deviations: the **states** 0, 1, 2, 3. More bins keep more detail, but need more samples to be counted well (slide 14).
3. Probability of a state = fraction of the 1200 volumes in that state: 0.15, 0.35, 0.34, 0.15 over all regions; more than 0.10 in every region.



Notebook, Part 1: the five regions of slide 2, and their four states.

Calling a fraction of time a probability assumes that the statistics of the signal do not change during the 14 minutes (stationarity).

A Gaussian signal cut at the same edges: 0.16, 0.34, 0.34, 0.16. Other choices of bins: notebook, Optional A

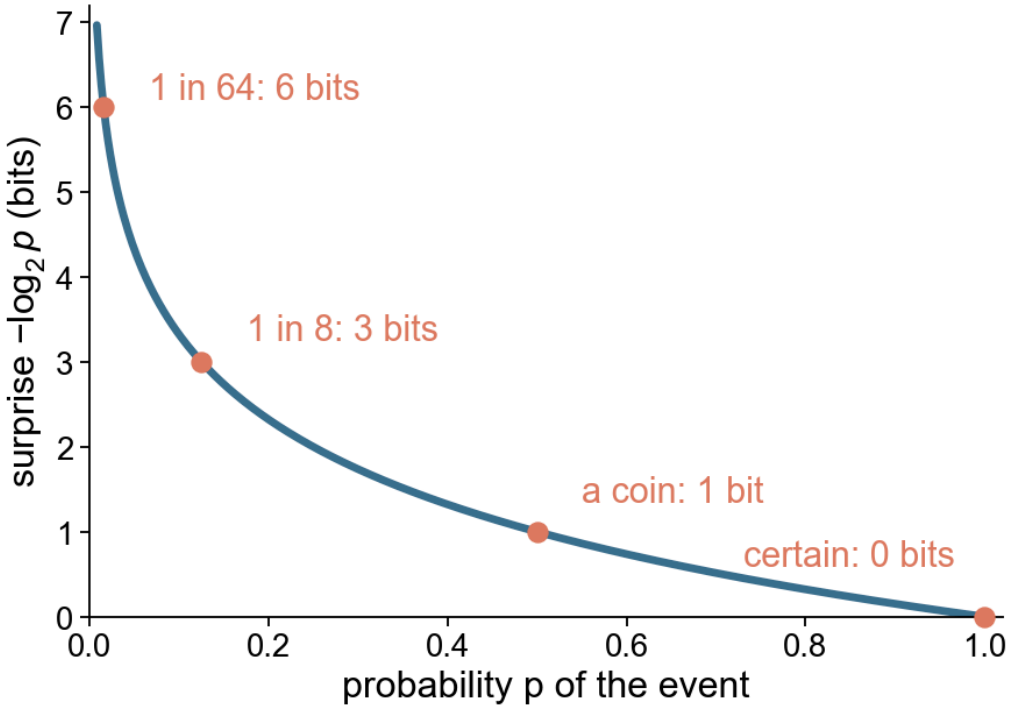
Surprise, from three requirements

How surprising is an event of probability p? Three requirements:

1. A certain event is no surprise: $S(1) = 0$.
2. Rarer events are more surprising: $S(p)$ decreases with p.
3. Two **independent** events, for which the probability of both is the product pq, have surprises that add: $S(pq) = S(p) + S(q)$.

From 3, $S(p^n) = n\,S(p)$; with 2, the only solutions are $S(p) = -c \log p$, $c > 0$. With $S(\tfrac{1}{2}) = \mathbf{1\ bit}$, one side of a fair coin:

$$S(p) = -\log_2 p = -\frac{\ln p}{\ln 2}$$



With the natural logarithm the unit is the nat: 1 nat = 1.44 bits.

Shannon (1948); the last step is the argument of his Appendix 2

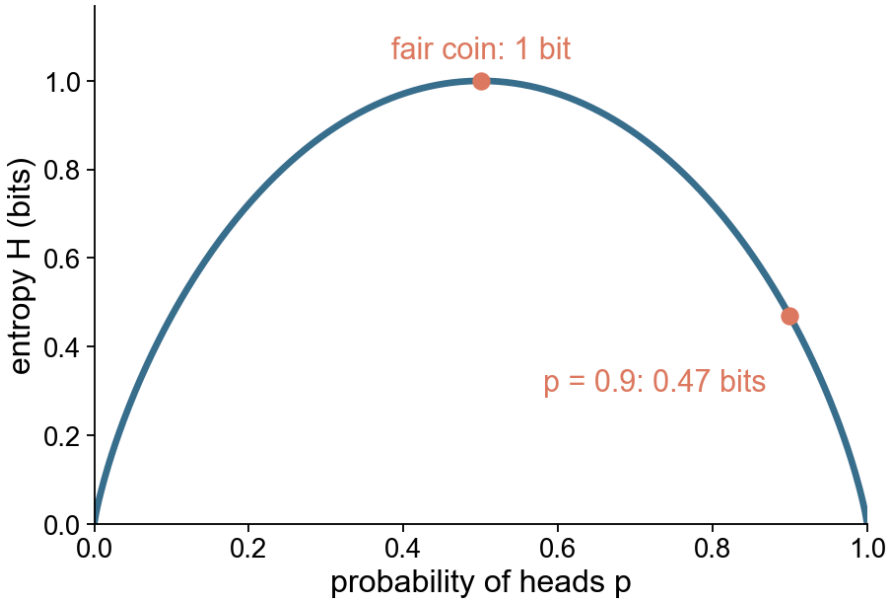
Entropy: the average surprise

Before we look, our uncertainty about the state of a signal X is the surprise we expect. X is in state x a fraction $p(x)$ of the time: averaging $-\log_2 p(x)$ with these weights gives the **entropy**,

$$H(X) = -\sum_x p(x)\,\log_2 p(x)$$

States that never occur do not count: $p \log p \to 0$ for $p \to 0$.

- Fair coin: $\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 1 = 1$ bit. A coin with $p = 0.9$: 0.47 bits, easier to guess.
- K equally likely states: $H = \log_2 K$, the largest value possible with K states (proof: slide 13).
- Region 0, four states: 1.866 of at most 2 bits.



A coin: the uncertainty is largest when heads and tails are equally likely.

Joint entropy

At each time point two regions X, Y are in one of **$4 \times 4 = 16$ joint states (x, y)**, with probability $p(x, y)$. Summing over y gives back $p(x)$, the **marginal**.

Two signals are **independent** when $p(x, y) = p(x)\,p(y)$ for every pair of states. The entropy of the pair is the **joint entropy**:

$$H(X, Y) = -\sum_{x,y} p(x, y)\, \log_2 p(x, y)$$

If regions 0 and 1 were independent, the surprises would add (slide 6) and $H(X_0, X_1)$ would be 1.866 + 1.876 = 3.742 bits. It is 3.640 bits: 0.102 bits less, because the regions depend on each other.

$x_0 \mid x_1$	*0*	*1*	*2*	*3*	$p(x_0)$
0	0.034	0.078	0.032	0.008	*0.152*
1	0.037	0.128	0.113	0.038	*0.317*
2	0.057	0.135	0.152	0.047	*0.391*
3	0.008	0.021	0.044	0.067	*0.140*
$p(x_1)$	*0.137*	*0.363*	*0.341*	*0.159*	*1*

Regions 0 and 1: the 16 joint probabilities, with the marginals in the margins. For independent regions the highlighted cell would be 0.140 × 0.159 = 0.022: the data have three times as much.

What is left: conditional entropy

1. Knowing $Y = y$, the probabilities of X become $p(x|y) = p(x,y)/p(y)$, with entropy $H(X|Y{=}y) = -\sum_x p(x|y)\log_2 p(x|y)$.
2. Averaged over y, with $p(y)\,p(x|y) = p(x,y)$, this is the **conditional entropy**, the uncertainty left on X once Y is known:

$$H(X|Y) = \sum_y p(y)\,H(X|Y{=}y) = -\sum_{x,y} p(x,y)\,\log_2 p(x|y)$$

3. Since $p(x,y) = p(y)\,p(x|y)$, $-\log_2 p(x,y) = -\log_2 p(y) - \log_2 p(x|y)$. Averaged over the joint states the three terms give $H(X,Y)$, $H(Y)$ (summing $p(x,y)$ over x leaves $p(y)$) and $H(X|Y)$: the **chain rule**

$$H(X,Y) = H(Y) + H(X|Y)$$

Regions 0 and 1

$H(X_0|X_1) = 3.640 - 1.876 = 1.764$ bits.

Knowing region 1 removes $1.866 - 1.764 = 0.102$ bits of uncertainty about region 0.

Lab A · 15 minutes

Parts 1 and 2 of the notebook

Part 1 · demonstration

The signals cut into four states, and the probabilities of the states: 0.15, 0.35, 0.34, 0.15.

Exercise 1 · entropy

From the counts of the states. Expected: $H(X_0) = 1.866$ bits; four equally likely states would give 2, a Gaussian signal cut in the same way 1.90.

Exercise 2 · conditional entropy

With the chain rule. Expected: $H(X_0|X_1) = 1.764$ and $H(X_1|X_0) = 1.774$ bits.

Question 1

Can four regions have less joint entropy than two?

How much does one signal tell about another? Mutual information

Cover & Thomas §2.3–2.8 and §8.1–8.6

Mutual information

The uncertainty on X removed by knowing Y:

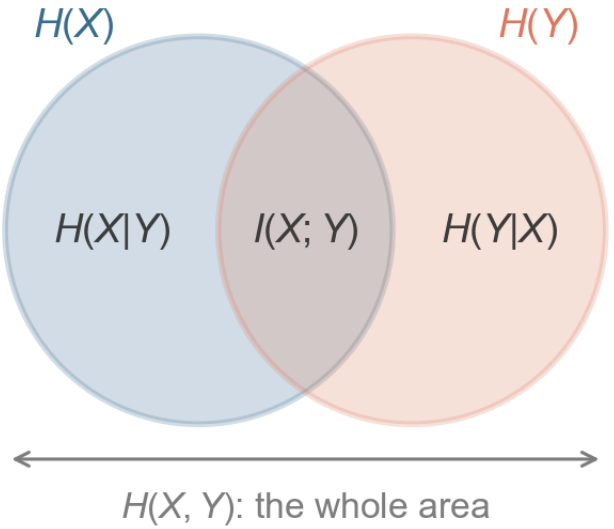
$$I(X;Y) = H(X) - H(X|Y)$$

With the chain rule, $H(X|Y) = H(X,Y) - H(Y)$, so

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

The expression is symmetric: Y tells as much about X as X tells about Y.

- Regions 0 and 1: 1.866 + 1.876 − 3.640 = 0.102 bits. Regions 4 and 5, with $\rho = 0.80$: 0.562 bits.
- $I(X;X) = H(X)$, because $H(X|X) = 0$: a signal tells everything about itself.



Zero only for independent signals

Write each entropy as an average over the joint states; for example $H(X) = -\sum_{x,y} p(x,y) \log_2 p(x)$, because $\sum_y p(x,y) = p(x)$. The three sums combine into one:

$$I(X;Y) = \sum_{x,y} p(x,y) \, \log_2 \frac{p(x,y)}{p(x)\,p(y)}$$

For independent signals (slide 8) every ratio is 1, every logarithm is 0, and $I = 0$.

The parabola and the circle of slide 3 have $\rho = 0$ and about 1 bit (0.97 and 1.17 bits, with 8 equally populated bins).

Why the logarithms cannot cancel, as the products did in ρ

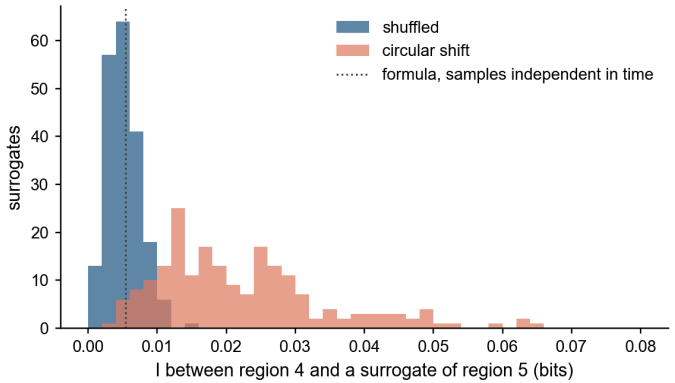
Since $\ln z \le z - 1$, with equality only at $z = 1$, summing over the joint states with $p(x,y) > 0$:

$$-I \ln 2 = \sum_{x,y} p(x,y) \ln \frac{p(x)p(y)}{p(x,y)} \le \sum_{x,y} \left[p(x)p(y) - p(x,y) \right] \le 0$$

The last step: over these states the $p(x,y)$ add up to 1, the $p(x)p(y)$ to at most 1. So $I \ge 0$, and $I = 0$ only if every ratio is 1: only for independent signals. The same inequality, applied to $p(x)$ and to $1/K$, gives $H \le \log_2 K$ (slide 7).

Counting has a bias

1. Counted probabilities fluctuate around the true ones. Since $I \geq 0$ for any table of counts (slide 13), the fluctuations cannot cancel: two signals independent of each other give $I > 0$ on average, a **bias**.
2. For T samples *independent in time* and $K_X \times K_Y$ states it is about $\frac{(K_X-1)(K_Y-1)}{2T \ln 2}$ bits (Miller 1955): 0.0054 bits here.
3. **Surrogates**, copies of region 5 with their dependence on region 4 removed on purpose, check it: shuffled in time, 0.005 bits.
4. BOLD is slow: correlation 0.98 between consecutive volumes, none after about 10 (7 s), so 1200 volumes hold about 300 independent samples. Rolled by at least 50 volumes, region 5 keeps its slowness and loses its alignment with region 4: 0.022 bits.



Notebook, Exercise 4: region 4 against 200 surrogates of region 5.

Rule

Compare every I with surrogates that keep the time course. 95% of the circular shifts stay below 0.05 bits; the real pair has 0.56.

Miller (1955). A circular shift removes the alignment, not slow drifts or rhythms shared by the two regions

Jointly Gaussian signals: I from ρ

1. One Gaussian: $-\ln p(x) = \frac{1}{2}\ln(2\pi\sigma^2) + \frac{(x-\mu)^2}{2\sigma^2}$. Average it, with $\langle (x-\mu)^2 \rangle = \sigma^2$, and write $\frac{1}{2} = \frac{1}{2}\ln e$:

$$h(X) = -\langle \ln p \rangle = \tfrac{1}{2}\ln\left(2\pi e\,\sigma^2\right)$$

2. Two jointly Gaussian signals, with covariance matrix $\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$: the same steps give $h(X,Y) = \frac{1}{2}\ln\left[(2\pi e)^2 \det\Sigma\right]$, with $\det\Sigma = \sigma_X^2\sigma_Y^2(1-\rho^2)$.

3. In $I = h(X) + h(Y) - h(X,Y)$ the factors $2\pi e$ and the variances cancel:

$$I(X;Y) = -\tfrac{1}{2}\ln\left(1-\rho^2\right)$$

in nats; divided by $\ln 2$, in bits. $\rho = 0.8$ gives 0.737 bits.

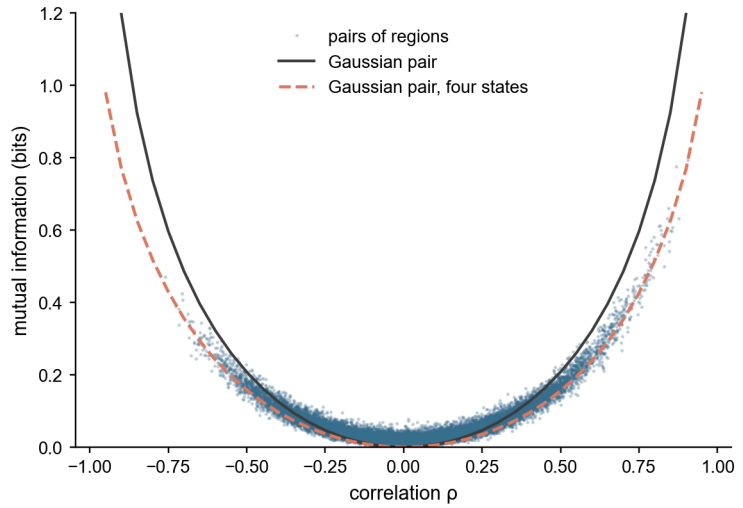
Why Gaussian

How much information can a correlation alone give? Take signals whose whole dependence is a correlation: jointly Gaussian signals, whose joint distribution is a two-dimensional Gaussian.

Continuous signals: with bins of width Δ, $H \approx h - \log\Delta$, with $h = -\int p \ln p\, dx$. H grows without limit as $\Delta \to 0$, and h depends on the units; in I the terms in Δ cancel, and I has a finite limit.

Pairs of regions are compatible with Gaussian pairs

1. Each point: one of the 19,900 pairs of regions, its counted I against its ρ. Black: the Gaussian pair of slide 15. Dashed: a Gaussian pair cut into four states, from 200,000 simulated samples, so without bias.
2. Dashed below black: a function of a signal cannot carry more information than the signal itself. At $\rho = 0.8$ cutting keeps 0.515 of 0.737 bits.
3. The pairs follow the dashed curve, on average 0.02 bits above it, about the bias of the circular shifts (slide 14). Weak pairs, with less information than the bias, lie above the black curve too.



Result

At this resolution, four states and 14 minutes, the counted I of the pairs is compatible with that of Gaussian pairs with the same ρ. It does not prove that the pairs are Gaussian.

Regions 4 and 5 ($\rho = 0.804$): 0.562 bits counted, 0.524 cut Gaussian. Data processing inequality: Cover & Thomas (2006), §2.8

Lab B · 15 minutes

Parts 3 and 4 of the notebook

Exercise 3 · mutual information

From three entropies. Expected: 0.562 bits for regions 4 and 5, 0.102 bits for regions 0 and 1. Then the three clouds of slide 3.

Exercise 4 · the bias

200 shuffles and 200 circular shifts of region 5. Expected: on average 0.005 and 0.022 bits.

Exercise 5 · a Gaussian pair

Write $-\frac{1}{2}\log_2(1-\rho^2)$. Expected: 0.737 bits at $\rho = 0.8$ (0.515 cut into four states), and the plot of slide 16.

Question 2

Where would the parabola appear on that plot? What does the plot say about a model fitted to the FC?

Can three signals share what no pair shows?

Redundancy and synergy · Rosas et al. (2019)

Copies and XOR

- **Copies:** one fair coin copied three times, $X = Y = Z$.
- **XOR:** two independent fair coins X, Y, and $Z = X \oplus Y$: 1 when they differ, 0 when they agree.

In the XOR every pair is independent: knowing X alone, Z is still a fair coin, because Y is. But the three are not independent together: Z is fixed by X and Y, so $p(x, y, z) \neq p(x)p(y)p(z)$.

A pair of signals is one signal with four states: $I(Z; X, Y) = H(Z) + H(X, Y) - H(X, Y, Z)$. XOR: $1 + 2 - 2 = 1$ bit.

	$I(Z;X)$	$I(Z;Y)$	$I(Z;X,Y)$
copies	1	1	1
XOR	0	0	1

Result

In both systems (X, Y) tells everything about Z. In the copies each signal alone tells it: **redundancy**. In the XOR only the two together: **synergy**, invisible to every pairwise correlation.

All values in bits; the two systems are drawn and computed in the notebook (Part 5)

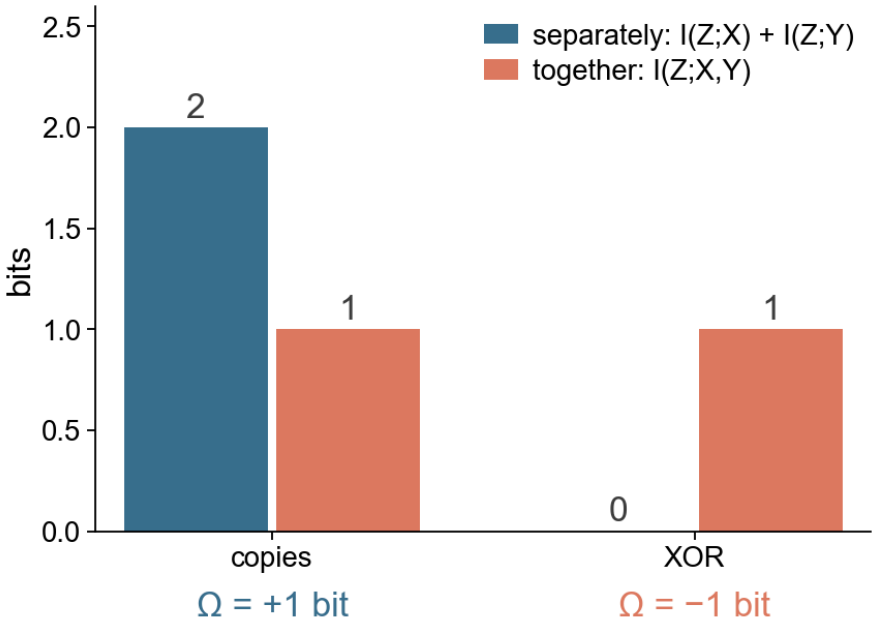
Which dominates: redundancy or synergy?

Choose one signal, the **target** Z, and compare what the other two tell about it separately and together:

$$\Omega = I(Z;X) + I(Z;Y) - I(Z;X,Y)$$

- Copies: $1 + 1 - 1 = +1$ bit. The second signal repeats the first: redundancy dominates, $\Omega > 0$.
- XOR: $0 + 0 - 1 = -1$ bit. The information appears only together: synergy dominates, $\Omega < 0$.

Ω is a balance: one triplet can hold both kinds, and the sign says which is larger. With entropies, $\Omega = H(X) + H(Y) + H(Z) - H(X,Y) - H(X,Z) - H(Y,Z) + H(X,Y,Z)$: exchanging the signals does not change it, so it does not depend on the choice of the target.



The O-information of Rosas et al. (2019), written for three signals. The entropy form: expand each I *as on slide 12*

Ω of three regions, without counting

1. Counting three regions means 64 joint states. With 1200 slow volumes, counting biases Ω by about −0.05 bits, more than $|\Omega|$ for most triplets.
2. We assume instead that three regions are jointly Gaussian: an assumption, since slide 16 tested pairs, and Gaussian pairs do not make a Gaussian triplet. Then $I(Z;X,Y) = -\frac{1}{2}\log_2(1-R^2)$, where R^2 is the fraction of the variance of Z explained by the best linear prediction from X and Y. For one partner $R^2 = \rho^2$: slide 15.
3. Least squares gives R^2 from the three correlations: $R^2 = \dfrac{\rho_{xz}^2 + \rho_{yz}^2 - 2\rho_{xy}\rho_{xz}\rho_{yz}}{1-\rho_{xy}^2}$. So, under this assumption, Ω of every triplet follows from the FC: all 1,313,400 triplets of regions in about a second.

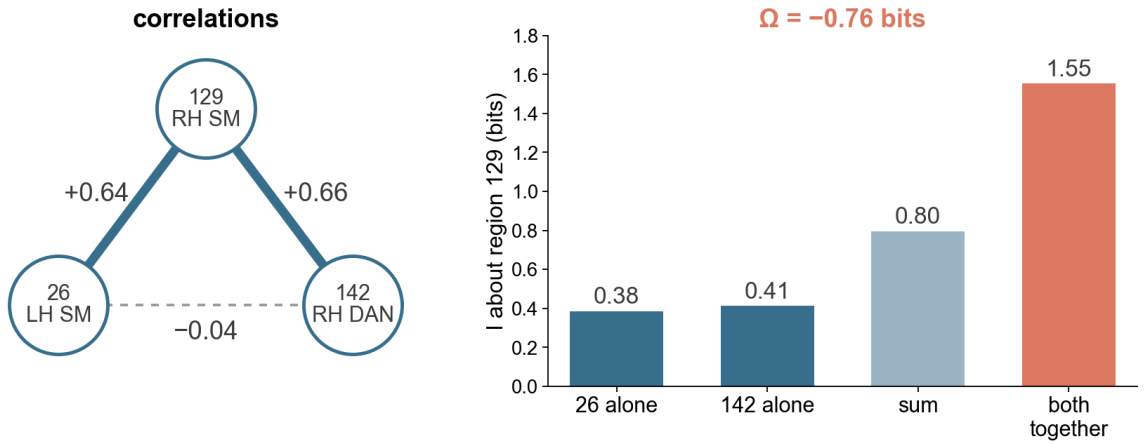
What this measures

The synergy of the Gaussian part of the dependence. It is in the FC, but only in three entries read together, never in one.

A synergy beyond the correlations, like the XOR's, escapes it.

Why $-\frac{1}{2}\log_2(1-R^2)$: for jointly Gaussian signals the variance of Z left after the prediction is $\sigma_Z^2(1-R^2)$, whatever values X and Y take.

The triplet with the most negative Ω



The most negative of the 1,313,400 estimates of this run, an extreme chosen among all of them. LH, RH: left, right hemisphere; SM: somatomotor; DAN: dorsal attention.

26 and 142 do not correlate (−0.04); both correlate with 129 (0.64, 0.66). About 129 they tell 0.384 and 0.412 bits alone, 1.553 together: $\Omega = 0.796 - 1.553 = -0.757$ bits.

Adding the bits of two partners means multiplying the fractions of variance they leave unexplained: $0.59 \times 0.56 = 0.33$. Together they leave only 0.12: 129 behaves like the sum of two independent signals.

Result

The FC shows two strong links and a missing one. Only the three entries read together show that the triplet holds more than its pairs.

Lab C · 13 minutes

Part 5 of the notebook

Part 5 · copies and XOR

What one signal and what a pair tell about Z. Expected: 1, 1 and 1 bit for the copies; 0, 0 and 1 bit for the XOR.

1.3 million triplets

Ω of every triplet of regions from its correlations, in about a second. Expected: $\Omega < 0$ in 47% of them, including all those with an odd number of negative correlations (36.5%); extremes −0.76 and +0.70 bits.

Exercise 6 · O-information

Separately minus together. Expected: +1.00 bits for the copies and −1.00 bits for the XOR, whichever signal is the target.

Question 3

Where does the extra information about region 129 come from? Would you notice this triplet in the FC?

Information in whole-brain models

- Pairs of resting BOLD signals are compatible with Gaussian pairs (slide 16): a model fitted to the FC is fitted to the pairwise information we can measure.
- Next lesson's model couples the regions through the connectome, the fibres between them seen by diffusion MRI. Where it fits the data it is linear and driven by Gaussian noise, so its fluctuations are Gaussian and its correlations fix today's quantities at zero lag.
- It misses dependence beyond the correlations, like the XOR, and, with connections that act equally both ways, identical regions and equal noise, any direction in time: Lesson 2 (Deco & Kringelbach, ch. 8).

Next lesson

Brain modeling: from the neuron to the whole brain. From a spiking neuron to 68 regions coupled through the connectome, with one global parameter fitted to the FC.

Four equations to take home

$$H(X) = -\sum_{x} p(x)\,\log_2 p(x)$$

Entropy is the average surprise. A region in four states: 1.87 of 2 bits.

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

Mutual information is zero only for independent signals. Compare it with surrogates that keep the time course.

$$I = -\tfrac{1}{2}\log_2\left(1-\rho^2\right)$$

Jointly Gaussian signals: the information depends on ρ alone. Pairs of BOLD signals are compatible with this at our resolution.

$$\Omega = I(Z;X) + I(Z;Y) - I(Z;X,Y)$$

Separately minus together: redundancy dominates if positive, synergy if negative. The XOR is invisible to every pair.

Reading and sources

Text. Cover TM, Thomas JA (2006). *Elements of Information Theory*, 2nd ed. Wiley. Chapter 2; §8.1–8.6 for continuous signals.

Deco G, Kringelbach ML (2026). *Whole-brain modelling: Cartography of the Dynamics of Mind*. Oxford University Press. doi:10.1093/oso/9780198991250.001.0001.

Luppi AI et al. (2022). A synergistic core for human brain evolution and cognition. *Nat Neurosci* 25:771–782.

Miller GA (1955). Note on the bias of information estimates. In: Quastler H (ed.), *Information Theory in Psychology*, Free Press, 95–100.

Rosas FE, Mediano PAM, Gastpar M, Jensen HJ (2019). Quantifying high-order interdependencies via multivariate extensions of the mutual information. *Phys Rev E* 100:032305.

Shannon CE (1948). A mathematical theory of communication. *Bell Syst Tech J* 27:379–423, 623–656.

Timme NM, Lapish C (2018). A tutorial for information theory in neuroscience. *eNeuro* 5:ENEURO.0052-18.2018.

Varley TF (2023). Information theory for complex systems scientists. arXiv:2304.12482.

Varley TF, Pope M, Faskowitz J, Sporns O (2023). Multivariate information theory uncovers synergistic subsystems of the human cerebral cortex. *Commun Biol* 6:451.

Data: WU-Minn Human Connectome Project (Van Essen et al. 2013, *NeuroImage* 80:62–79); Schaefer et al. (2018), *Cereb Cortex* 28:3095–3114.